

MUHAMMAD MAHEEM

AI Developer and Machine Learning Engineer

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EDUCATION

Superior University, Lahore — BS Artificial Intelligence

2024 – 2028

EXPERIENCE

AI Engineer Intern

Jun 2026 – Present | Remote

TEYZIX CORE

- Built **TEYZIX**, an AI-powered document summarizer combining extractive scoring (frequency, TF-IDF, positional ranking) with abstractive BART (facebook/bart-large-cnn).
- Added multi-document support and auto-detection of 16+ languages with multilingual stopword handling, plus PDF/TXT upload and export.

AI Developer Intern

Mar 2026 – May 2026 | Paid | Remote

Punjab Skill Development Authority (PSDA)

- Sole engineer delivering a full-stack SaaS platform (Next.js + Tailwind + PostgreSQL) from Figma specs to production — public site, admin dashboard, and a 3-role permission system across 8+ REST API endpoints.
- Built an OCR document parser using the Gemini API to extract and structure text from multi-page uploads, integrated into the SaaS backend pipeline.

Software Engineer Intern

Sep 2025 – Jan 2026 | Part-time | Remote

Artificial Automation Business Solutions (AABS)

- Built a RAG-based PDF chatbot with embedding-model selection, ChromaDB vector indexing, and retrieval-augmented generation, improving answer grounding over a naive keyword-search baseline.
- Developed responsive SaaS dashboard UI in Next.js 15 / React 19 / TypeScript, including authentication, analytics visualizations, and subscription-management interfaces.

PROJECTS

Quality Evaluation PDF Chatbot — Production RAG pipeline with a hard RAGAS ≥ 0.80 quality gate for CI/CD, hybrid search (dense + BM25 + reranking), and Prometheus observability. *Python, FastAPI, ChromaDB, Groq/Llama 3, LangGraph.*

Agentic RAG PDF Chatbot — Full-stack PDF chatbot with agentic multi-query RAG, SSE token streaming, JWT per-user workspaces, and page-level citations. *FastAPI, Next.js, Groq, pgvector.*

Hadith Chatbot — 39,038 authentic Hadiths with a two-stage BM25 → FAISS semantic re-ranking pipeline and LangChain orchestration for precise cross-hadith attribution.

Tic-Tac-Toe AI (DQN) — PyTorch reinforcement-learning agent trained over 100k self-play episodes to a >98% win/draw rate, benchmarked against a minimax solver.

KRR Application — AI research platform: PDF ingestion, structured summarization, cross-paper analysis, and pgvector semantic search. *FastAPI, Next.js, Groq, Docker.*

CKD Prediction Model — Streamlit app predicting Chronic Kidney Disease from 12 clinical features (Random Forest, 96% accuracy, 1,659 patients).

TECHNICAL SKILLS

AI & Machine

Python • PyTorch • Scikit-learn • OpenCV • RAG Systems • LangChain • LangGraph •

Learning

ChromaDB • FAISS • Sentence Transformers • Groq API • Gemini API • OpenAI API • LLM Evaluation (RAGAS) • Multi-Agent Systems • Reinforcement Learning • Prompt Engineering

Web & Backend

Next.js • React • FastAPI • TypeScript • JavaScript • Tailwind CSS • PostgreSQL • Firebase • Streamlit • REST APIs

Tools

Git / GitHub • Linux (Kali) • Docker (basics) • Claude Code • Copilot CLI • CI/CD Integration • Data Structures